

Experiment No: _

Date: _____

Two ray channel, Okumura and Hata model

Aim: To evaluate median path loss for Two- Ray, Okumura model and Hata model for outdoor propagation.

Tools Required: MATLAB R2015a/ Simulink library

Theory: The Okumura model for Urban Areas is a Radio propagation model that was built using data. mura Model. Okumura's model is one of the most widely used models for signal prediction in urban areas. This model is applicable for frequencies in the range 150MHz to 1920 MHz (although it is typically extrapolated up to 3000 MHz) and distances of 1 km to 100 km. It can be used for base station antenna heights ranging from 30 m to 1000 m.

Okumura model was built into three modes: the ones for urban, suburban and open areas. The model for urban areas was built first and used as the base for others. Okumura developed a set of curves giving the median attenuation relative to free space (A_{mu}), in an urban area over a quasi-smooth terrain with a base station effective antenna height (h_{te}) of 200 m and a mobile antenna height (h_{re}) of 3 m.

Coverage

Frequency = 150 MHz to 1920 MHz

Mobile Station Antenna Height: between 1 m and 10 m

Base station Antenna Height: between 30 m and 1000 m

Link distance: between 1 km and 100 km

Hata Model:

The Hata model is an empirical formulation of the graphical path loss data provided by Okumura, and is valid from 150 MHz to 1500 MHz. Hata presented the urban area propagation loss as a standard formula and supplied correction equations for application to other situations. Hata Model predicts the total path loss along a link of terrestrial microwave or other type of cellular communications. This particular version of the Hata model is applicable to the radio propagation within urban areas. This model is suited for both point-to-point and broadcast transmissions and it is based on extensive empirical measurements taken.

Hata Model for Urban Areas is formulated as:

$$L_U = 69.55 + 26.16 \log f - 13.82 \log h_B - CH + [44.9 - 6.55 \log h_B] \log d.$$

For small or medium sized city,

$$CH = 0.8 + (1.1 \log f - 0.7) h_M - 1.56 \log f.$$

and for large cities,

$$CH = 8.29 (\log (1.54 h_M))^2 - 1.1, \text{ if } 150 \leq f \leq 200$$

$$CH = 3.2 (\log (11.75 h_M))^2 - 4.97, \text{ if } 200 \leq f \leq 1500$$

Where,

L_U = Path loss in Urban Areas (dB)

h_B = Height of base station Antenna. (m)

h_M = Height of mobile station Antenna. (m)

f= Frequency of Transmission (MHz).

CH = Antenna height correction factor

d= Distance between the base and mobile stations (km).

The term "small city" means a city where the mobile antenna height not more than 10 meters. i.e.

$1 \leq h_m \leq 10\text{m}$

Two ray Ground reflection model-

$$PL(\text{dB}) = 40\log d - (10\log G_t + 10\log G_r + 20\log h_t + 20\log h_r)$$

$$P_r = P_t G_t G_r \frac{h_t^2 h_r^2}{d^4}$$

MATLAB Program:

```
clc;
```

```
clear;
```

```
close all;
```

```
%Enter Input Parameters%
```

```
f = input('Enter frequency (MHz): '); % 150–1500 MHz
```

```
hb = input('Enter base station height (m): '); % 30–200 m
```

```
hm = input('Enter mobile height (m): '); % 1–10 m
```

```
% Distance range (km)
```

```
d = 1:0.5:20;
```

```
% Convert frequency to Hz for two-ray model
```

```
f_hz = f * 1e6;
```

```
c = 3e8;
```

```
lambda = c / f_hz;
```

```
if f < 150 || f > 1500
```

```
    error('Frequency out of valid range (150–1500 MHz)');
```

```
end
```

```
if hb < 30 || hb > 200
```

```
    error('Base station height out of range (30–200 m)');
```

```
end
```

```

if hm < 1 || hm > 10
    error('Mobile height out of range (1-10 m)');
end

%Initilize path loss for all three models
PL_two_ray = zeros(size(d));
PL_okumura = zeros(size(d));
PL_hata = zeros(size(d));
for i = 1:length(d)
    dist_km = d(i);
    dist_m = dist_km * 1000;

    %Ground Two-Ray Model
    if dist_m < 10*sqrt(hb*hm)
        % Not strictly valid, still computed
        warning('Distance may not satisfy two-ray condition');
    end

    Pr_Pt = (hb^2 * hm^2) / (dist_m^4);
    PL_two_ray(i) = -10*log10(Pr_Pt); % Path loss in dB

    % Okumura Model
    Lf = 32.45 + 20*log10(f) + 20*log10(dist_km);
    Amu = 35 + 20*log10(dist_km); % Approximation
    Ghb = 20*log10(hb/200);
    Ghm = 10*log10(hm/3);
    Garea = 0; % Urban

    PL_okumura(i) = Lf + Amu - Ghb - Ghm - Garea;

    % Hata Model (Urban)
    ch = 0.8 + (1.1*log10(f)-0.7)*hm - 1.56*log10(f);

    PL_hata(i) = 69.55 + 26.16*log10(f) ...
        - 13.82*log10(hb) - ch ...
        + (44.9 - 6.55*log10(hb))*log10(dist_km);
end

```

```
end  
  
figure;  
  
plot(d, PL_two_ray, 'LineWidth', 2); hold on;  
  
plot(d, PL_okumura, 'LineWidth', 2);  
  
plot(d, PL_hata, 'LineWidth', 2);  
  
grid on;  
  
xlabel('Distance (km)');  
  
ylabel('Path Loss (dB)');  
  
title('Path Loss Comparison: Ground Two-Ray vs Okumura vs Hata');  
  
legend('Ground Two-Ray', 'Okumura', 'Hata (Urban)');
```

